

YouTube Trend Analysis Project Documentation

Project Title

YouTube Trend Analysis Dashboard using SQL and Power BI

1. Project Overview

This project focuses on analyzing YouTube trending video data to identify audience engagement patterns, high-performing content categories, hashtag effectiveness, publishing trends, and channel growth behavior.

The project was built using:

- SQL for data analysis
- Power BI for dashboard creation and storytelling
- ERD-based relational data modeling

The goal of the project is to help content creators and businesses understand:

- Which type of content performs best
 - What drives engagement
 - Which hashtags and categories attract attention
 - Which publishing patterns support growth
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2. Business Problem

Content creators and digital businesses often struggle to understand:

- What type of videos generate the highest audience engagement
- Which categories receive the most views
- How hashtags influence content visibility
- Which upload patterns support channel growth
- What characteristics are common among fast-growing videos

The objective of this project is to analyze YouTube trend data and provide actionable insights that can help creators improve audience reach and engagement.

3. Dataset Description

The dataset contains YouTube-related information including:

- Videos
- Channels

- Categories
- Hashtags
- Video performance snapshots

The dataset includes over 1000+ records across multiple tables.

4. Database Schema

The project follows a relational database structure.

Main Tables:

Table Name	Description
videos	Stores video details
channels	Stores channel information
categories	Stores video categories
hashtags	Stores hashtag names
video_snapshots	Stores views, likes, comments over time
video_categories	Bridge table between videos and categories
video_hashtags	Bridge table between videos and hashtags

5. ERD Design

The ERD was designed using bridge tables to handle many-to-many relationships.

Relationships:

- One channel can upload many videos
 - One video can belong to multiple hashtags
 - One video can belong to multiple categories
 - One video can have multiple performance snapshots
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6. Data Cleaning & Validation

The following validation checks were performed:

Checks Performed

- Counted total records in each table

- Checked NULL values in important columns
- Verified relationships between tables
- Validated hashtag filtering logic in Power BI
- Fixed ambiguous relationship paths in the data model

Findings

- No NULL values were found in important fields
 - All relationships were successfully validated
 - Data model issues were resolved using proper relationship directions
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7. SQL Analysis Performed

The following analyses were performed using SQL:

Video Performance Analysis

- Top videos by views
- Top videos by likes
- Top videos by comments
- Growth analysis using max views and min views

Engagement Analysis

- Likes to views ratio
- Comments to views ratio

Category Analysis

- Total videos per category
- Average views by category
- Average likes by category

Hashtag Analysis

- Total videos per hashtag
- Average views by hashtag

Channel Analysis

- Total videos per channel
- Average views per channel
- Average likes per channel

Publishing Trend Analysis

- Day-wise upload analysis
 - Average views by publishing day
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8. Power BI Dashboard

The dashboard was divided into two pages:

Page 1: Overview Dashboard

The overview page provides a high-level summary of YouTube trends.

Visuals Included

- KPI Cards
 - Total Videos
 - Total Views
 - Average Views
 - Engagement Ratio
 - Average Views by Category
 - Top Performing Hashtags
 - Key Insight Summary
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Page 2: Deep Dive Analysis

The deep dive page focuses on audience engagement and growth behavior.

Visuals Included

- Views vs Likes Scatter Plot
 - Channel Performance Analysis
 - Growth Analysis
 - Best Publishing Day Analysis
 - Key Insight Panel
 - Interactive Filters
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9. Key Insights

Category Insights

- Music category generated the highest average views
- Comedy and entertainment content also performed strongly

- Education and science-related content received comparatively lower views

Engagement Insights

- Videos with emotional and humorous content generated stronger engagement
- Some videos gained high views but lower likes, indicating passive audience behavior

Hashtag Insights

- Comedy and entertainment hashtags showed strong audience reach
- Videos using multiple hashtags received better visibility

Growth Insights

- Short-form and nostalgic content showed the fastest growth
- Cartoon and comedy-based videos demonstrated strong viral potential

Channel Insights

- Channels with consistent upload frequency performed better overall
- Entertainment-focused channels dominated total views

Publishing Insights

- Tuesday showed the highest average views in the dataset
- However, the lower upload volume suggests the result may be influenced by viral outliers

10. Business Recommendations

Based on the analysis:

- Creators should focus on short-form entertainment content
- Comedy and emotionally engaging storytelling can improve audience retention
- Consistent uploads are important for channel growth
- Hashtag optimization can improve content discoverability
- Upload strategies should be tested across multiple days instead of relying on one trend

11. Challenges Faced

During the project, the following challenges were encountered:

- Ambiguous relationships in Power BI
- Incorrect hashtag filtering behavior
- Day-wise sorting issues in Power BI visuals
- Data model debugging and filter propagation issues

These challenges were resolved by:

- Fixing relationship directions
 - Using bridge table logic properly
 - Applying correct DAX measures
 - Sorting day names using day number columns
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12. Skills Demonstrated

SQL Skills

- Joins
- Aggregations
- GROUP BY
- HAVING
- Ratio calculations
- Analytical querying

Power BI Skills

- Data modeling
- Relationship management
- DAX measures
- Dashboard design
- Visual storytelling
- Interactive filtering

Analytical Skills

- Business insight generation
 - Trend analysis
 - Engagement analysis
 - Performance evaluation
 - Data validation and debugging
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13. Tools & Technologies Used

Tool	Purpose
SQL	Data analysis
Power BI	Dashboard creation
ERD Tool	Database modeling

14. Conclusion

This project successfully analyzed YouTube trend data to identify patterns in audience engagement, content performance, and publishing behavior.

The dashboard provides actionable business insights that can help creators and businesses optimize content strategy, improve engagement, and increase audience reach.

The project also strengthened practical skills in SQL, Power BI, data modeling, dashboard storytelling, and business analysis.